

1. The inequality constraints are

$$\begin{aligned} -x_1 + x_2 &\leq 2, \\ x_1 + x_2 &\leq 4, \\ -x_1 &\leq 0, \\ -x_2 &\leq 0. \end{aligned}$$

2. Each of the four constraints defines a half-plane, and the intersection of the half-planes gives the polyhedron in $\mathbb{R}^2$. We draw first the lines defined by the equality version of the constraints:

$$\begin{aligned} -x_1 + x_2 &= 2, \\ x_1 + x_2 &= 4, \\ -x_1 &= 0, \\ -x_2 &= 0. \end{aligned}$$

In order to identify on which side of the line is the half-plane, we use the gradient with the opposite sign. Indeed, the gradient of a function indicates the direction in which the function increases. Here, as the function must be less or equal to a constant, we need the direction in which the function decreases.

For instance, for the first constraint, the gradient is the first row of the matrix A . Written as a column vector, it is:

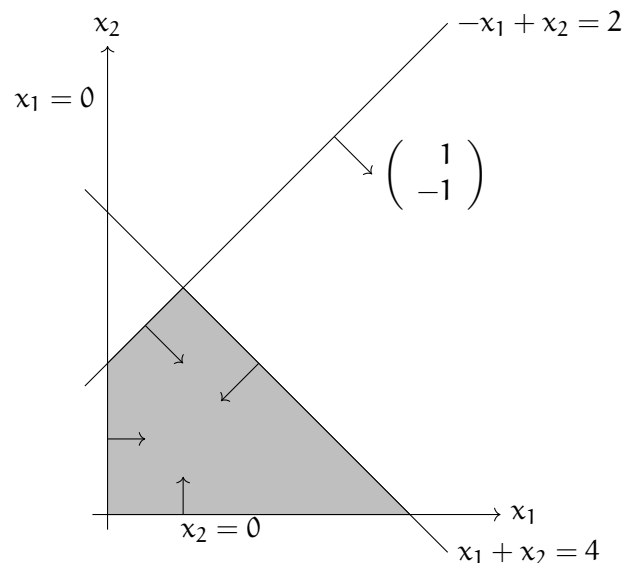
$$\begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Therefore, the direction that shows which side of the line is feasible is

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The other directions are obtained in the same way:

$$d_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, d_2 = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, d_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, d_4 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$



3. In order to write the standard form of the polyhedron, we need to include the slack variables. As there are 4 inequality constraints, we need 4 slack variables:

$$\mathcal{P} = \{x \in \mathbb{R}^2, x^s \in \mathbb{R}^4 \mid Ax + x^s = b, x^s \geq 0\}. \quad (1)$$

We now have 6 variables. The set of constraints characterizing the polyhedron is:

$$\begin{array}{rcccccccl} -x_1 & +x_2 & +x_3 & & & & & = 2, \\ x_1 & +x_2 & & +x_4 & & & & = 4, \\ -x_1 & & & & +x_5 & & & = 0, \\ & -x_2 & & & & +x_6 & & = 0, \\ x_1 & & & & & & & \geq 0, \\ & x_2 & & & & & & \geq 0, \\ & & x_3 & & & & & \geq 0, \\ & & & x_4 & & & & \geq 0, \\ & & & & x_5 & & & \geq 0, \\ & & & & & x_6 & & \geq 0. \end{array}$$